

2.29 Numerical Fluid Mechanics

Spring 2015 – Lecture 16

REVIEW Lecture 15:

- Finite Volume Methods

- Integral and conservative forms of the cons. laws
- Introduction
- Approximations needed and basic elements of a FV scheme
 - Grid generation $\Rightarrow$ Time-Marching
 - FV grids: Cell centered (Nodes or CV-faces) vs. Cell vertex; Structured vs. Unstructured
 - Approximation of surface integrals (leading to symbolic formulas)
 - Approximation of volume integrals (leading to symbolic formulas)
 - Summary: Steps to step-up a FV scheme
- One Dimensional examples
 - Generic equation:
$$\frac{d(\Delta x \bar{\Phi}_j)}{dt} + f_{j+1/2} - f_{j-1/2} = \int_{x_{j-1/2}}^{x_{j+1/2}} s_\phi(x, t) dx$$
 - Linear Convection (Sommerfeld eqn): convective fluxes
 - 2nd order in space

TODAY (Lecture 16): FINITE VOLUME METHODS

- Summary: Steps to step-up a FV scheme
- Examples: One Dimensional examples
 - Generic equations
 - Linear Convection (Sommerfeld eqn): convective fluxes
 - 2nd order in space, 4th order in space, links to CDS
 - Unsteady Diffusion equation: diffusive fluxes
 - Two approaches for 2nd order in space, links to CDS
- Approximation of surface integrals and volume integrals revisited
- Interpolations and differentiations
 - Upwind interpolation (UDS)
 - Linear Interpolation (CDS)
 - Quadratic Upwind interpolation (QUICK)
 - Higher order (interpolation) schemes

References and Reading Assignments

- Chapter 29.4 on “The control-volume approach for elliptic equations” of “Chapra and Canale, Numerical Methods for Engineers, 2014/2010/2006.”
- Chapter 4 on “Finite Volume Methods” of “J. H. Ferziger and M. Peric, Computational Methods for Fluid Dynamics. Springer, NY, 3rd edition, 2002”
- Chapter 5 on “Finite Volume Methods” of “H. Lomax, T. H. Pulliam, D.W. Zingg, *Fundamentals of Computational Fluid Dynamics (Scientific Computation)*. Springer, 2003”
- Chapter 5.6 on “Finite-Volume Methods” of T. Cebeci, J. P. Shao, F. Kafyeke and E. Laurendeau, Computational Fluid Dynamics for Engineers. Springer, 2005.

One-Dimensional Example I

Linear Convection (Sommerfeld) Eqn, Cont'd

- The resultant linear algebraic system is circulant tri-diagonal (for periodic BCs)

$$\frac{d \bar{\Phi}}{dt} + \frac{c}{2\Delta x} \mathbf{B}_p(-1, 0, 1) \bar{\Phi} = 0$$

- This is as the 2nd order CDS!, except that it is written in terms of cell averaged values instead of values at FD nodes/points
 - It is also 2nd order in space
 - Has same properties as classic CDS for $\frac{\partial \phi(x, t)}{\partial t} + \frac{\partial c \phi(x, t)}{\partial x} = 0$
 - Non-dissipative (check Fourier analysis or eigenvalues of $\mathbf{B}_p$ which are imaginary), but can provide oscillatory errors
 - Stability (recall tables for FD schemes, linear convection eqn.) of time-marching
 - If centered in time, centered in space, explicit: stable with CFL condition: $\frac{c \Delta t}{\Delta x} \leq 1$
 - If implicit in time: unconditionally stable for all $\Delta t, \Delta x$

One-Dimensional Example II

Linear Convection (Sommerfeld) Eqn: 4th order approx.

- 1D exact integral equation still

$$\frac{d(\Delta x \bar{\Phi}_j)}{dt} + f_{j+1/2} - f_{j-1/2} = 0$$

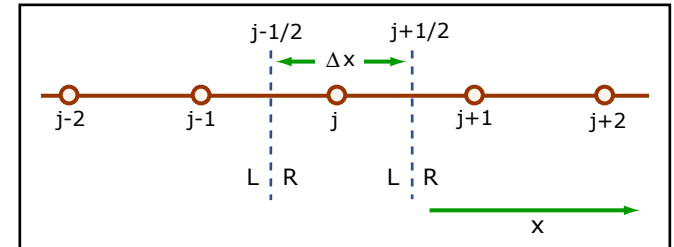


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- Use 4th order accurate surface/volume integrals

- Replace piecewise-constant approx. to $\phi(x)$ with piece-wise quadratic approx ($\xi = x - x_j$): $\phi(\xi) = a\xi^2 + b\xi + c$ (note ϕ defined over more than 1 cell)
- Satisfy $\bar{\Phi}_p$'s (average) constraints, i.e. choose a, b, c so that:

$$\frac{1}{\Delta x} \int_{-3\Delta x/2}^{-\Delta x/2} \phi(\xi) d\xi = \bar{\phi}_{j-1}, \quad \frac{1}{\Delta x} \int_{-\Delta x/2}^{+\Delta x/2} \phi(\xi) d\xi = \bar{\phi}_j, \quad \frac{1}{\Delta x} \int_{\Delta x/2}^{3\Delta x/2} \phi(\xi) d\xi = \bar{\phi}_{j+1}$$

- This gives:

$$a = \frac{\bar{\phi}_{j+1} - 2\bar{\phi}_j + \bar{\phi}_{j-1}}{2\Delta x^2}, \quad b = \frac{\bar{\phi}_{j+1} - \bar{\phi}_{j-1}}{2\Delta x}, \quad c = \frac{-\bar{\phi}_{j-1} + 26\bar{\phi}_j - \bar{\phi}_{j+1}}{24}$$

- Next, we need to evaluate the values of $\phi(x)$ at the boundaries so as to compute the advective fluxes at these boundaries: $f_{j-1/2}^L, f_{j-1/2}^R, f_{j+1/2}^L, f_{j+1/2}^R$

One-Dimensional Example II

Linear Convection (Sommerfeld) Eqn: 4th order approx.

- Since $f = c\phi \Rightarrow$ compute ϕ at edges:

$$\phi_{j-1/2}^L = \frac{2\bar{\phi}_j + 5\bar{\phi}_{j-1} - \bar{\phi}_{j-2}}{6}, \quad \phi_{j+1/2}^L = \frac{2\bar{\phi}_{j+1} + 5\bar{\phi}_j - \bar{\phi}_{j-1}}{6},$$

$$\phi_{j-1/2}^R = \frac{-\bar{\phi}_{j+1} + 5\bar{\phi}_j + 2\bar{\phi}_{j-1}}{6}, \quad \phi_{j+1/2}^R = \frac{-\bar{\phi}_{j+2} + 5\bar{\phi}_{j+1} + 2\bar{\phi}_j}{6}$$

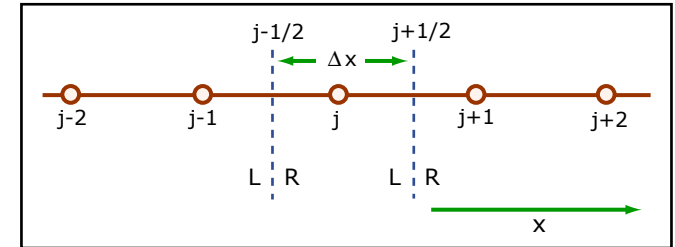


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- Resolve flux discontinuity $\Rightarrow$ again, use average values

$$\hat{f}_{j-1/2} = \frac{f_{j-1/2}^L + f_{j-1/2}^R}{2} = \frac{c\phi_{j-1/2}^L + c\phi_{j-1/2}^R}{2}$$

$$\Rightarrow \hat{f}_{j-1/2} = c \frac{-\bar{\phi}_{j+1} + 7\bar{\phi}_j + 7\bar{\phi}_{j-1} - \bar{\phi}_{j-2}}{12}$$

$$\hat{f}_{j+1/2} = \frac{f_{j+1/2}^L + f_{j+1/2}^R}{2} = \frac{c\phi_{j+1/2}^L + c\phi_{j+1/2}^R}{2}$$

$$\Rightarrow \hat{f}_{j+1/2} = c \frac{-\bar{\phi}_{j+2} + 7\bar{\phi}_{j+1} + 7\bar{\phi}_j - \bar{\phi}_{j-1}}{12}$$

- Done with “integrals” $\Rightarrow$ we can substitute in 1D conv. eqn:

$$\frac{d(\Delta x \bar{\Phi}_j)}{dt} + f_{j+1/2} - f_{j-1/2} \approx \frac{d(\Delta x \bar{\phi}_j)}{dt} + \hat{f}_{j+1/2} - \hat{f}_{j-1/2} \Rightarrow \Delta x \frac{d\bar{\phi}_j}{dt} + c \frac{-\bar{\phi}_{j+2} + 8\bar{\phi}_{j+1} - 8\bar{\phi}_{j-1} + \bar{\phi}_{j-2}}{12} = 0$$

- For periodic domains:

$$\frac{d\bar{\Phi}}{dt} + \frac{c}{12\Delta x} \mathbf{B}_P(-1, -8, 0, 8, 1) \bar{\Phi} = 0$$

3.3

Centered finite-divided-difference formulas: two versions are presented for each derivative. The latter version incorporates more terms of the Taylor series expansion and is, consequently, more accurate.

Centered Differences

<u>First Derivative</u>		<u>Error</u>
$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h}$		$O(h^2)$
$f'(x_i) = \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2}))}{12h}$		$O(h^4)$
<u>Second Derivative</u>		
$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2}$		$O(h^2)$
$f''(x_i) = \frac{-f(x_{i+2}) + 16f(x_{i+1}) - 30f(x_i) + 16f(x_{i-1}) - f(x_{i-2}))}{12h^2}$		$O(h^4)$
<u>Third Derivative</u>		
$f'''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + 2f(x_{i-1}) - f(x_{i-2}))}{2h^3}$		$O(h^2)$
$f'''(x_i) = \frac{-f(x_{i+3}) + 8f(x_{i+2}) - 13f(x_{i+1}) + 13f(x_{i-1}) - 8f(x_{i-2}) + f(x_{i-3}))}{8h^3}$		$O(h^4)$
<u>Fourth Derivative</u>		
$f^{(4)}(x_i) = \frac{f(x_{i+2}) - 4f(x_{i+1}) + 6f(x_i) - 4f(x_{i-1}) + f(x_{i-2}))}{h^4}$		$O(h^2)$
$f^{(4)}(x_i) = \frac{-f(x_{i+3}) + 12f(x_{i+2}) + 39f(x_{i+1}) + 56f(x_i) - 39f(x_{i-1}) + 12f(x_{i-2}) + f(x_{i-3}))}{6h^4}$		$O(h^4)$

One-Dimensional Example III

2nd order approx. of diffusion equation:

$$\frac{\partial \phi(x,t)}{\partial t} = \nu \frac{\partial^2 \phi(x,t)}{\partial x^2}$$

- 1D exact integral equation same form!

$$\frac{d(\Delta x \bar{\Phi}_j)}{dt} + f_{j+1/2} - f_{j-1/2} = 0$$

but with: $f = -\nu \nabla \phi = -\nu \frac{\partial \phi}{\partial x}$

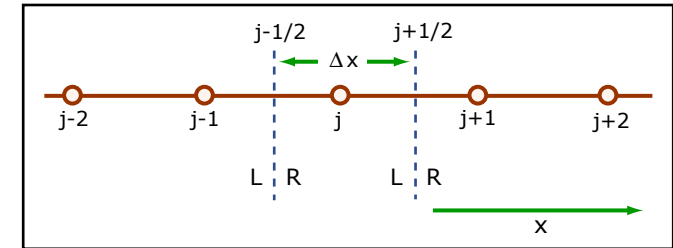


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- Approximation of surface (flux) integral: Approach 1

– Direct: we know that to second-order (from CDS and from $\bar{\phi}_j = \phi_j + O(\Delta x^2)$)

$$f_{j+1/2} = -\nu \frac{\partial \phi}{\partial x} \Big|_{j+1/2} = -\nu \frac{\bar{\phi}_{j+1} - \bar{\phi}_j}{\Delta x} + O(\Delta x^2) \quad \Rightarrow \quad \hat{f}_{j+1/2} = -\nu \frac{\bar{\phi}_{j+1} - \bar{\phi}_j}{\Delta x} \quad \text{and} \quad \hat{f}_{j-1/2} = -\nu \frac{\bar{\phi}_j - \bar{\phi}_{j-1}}{\Delta x}$$

– Substitute into integral equation:

$$\frac{d(\Delta x \bar{\phi}_j)}{dt} + \hat{f}_{j+1/2} - \hat{f}_{j-1/2} = \Delta x \frac{d \bar{\phi}_j}{dt} + \nu \frac{\bar{\phi}_{j-1} - 2\bar{\phi}_j + \bar{\phi}_{j+1}}{\Delta x} = 0$$

– In the matrix form, with Dirichlet BCs:

- Semi-discrete FV scheme is as CDS in space,
but in terms of cell-averaged data

$$\frac{d \bar{\Phi}}{dt} = \frac{\nu}{\Delta x^2} \mathbf{B}(1, -2, 1) \bar{\Phi} + (\mathbf{bc})$$

One-Dimensional Example III

2nd order approx. of diffusion equation:

$$\frac{\partial \phi(x,t)}{\partial t} = \nu \frac{\partial^2 \phi(x,t)}{\partial x^2}$$

• Approximation of surface (flux) integral: Approach 2

– Use a piece-wise quadratic approx.: $\phi(\xi) = a\xi^2 + b\xi + c$ $\Rightarrow \frac{\partial \phi}{\partial x} = \frac{\partial \phi}{\partial \xi} = 2a\xi + b$

• Note that a, b, c remain as before, they are set by the volume average constraints

• Since a, b are “symmetric”: $f_{j+1/2}^R = f_{j+1/2}^L = -\nu \frac{\partial \phi}{\partial x} \Big|_{j+1/2} = -\nu \frac{\bar{\phi}_{j+1} - \bar{\phi}_j}{\Delta x} + O(\Delta x^2)$

$$f_{j-1/2}^R = f_{j-1/2}^L = -\nu \frac{\partial \phi}{\partial x} \Big|_{j-1/2} = -\nu \frac{\bar{\phi}_j - \bar{\phi}_{j-1}}{\Delta x} + O(\Delta x^2)$$

• There are no flux discontinuities in this case

– Substitute into integral equation:

$$\frac{d(\Delta x \bar{\phi}_j)}{dt} + \hat{f}_{j+1/2} - \hat{f}_{j-1/2} = \Delta x \frac{d \bar{\phi}_j}{dt} + \nu \frac{\bar{\phi}_{j-1} - 2\bar{\phi}_j + \bar{\phi}_{j+1}}{\Delta x} = 0$$

– In the matrix form, with Dirichlet BCs:

• Semi-discrete FV scheme is as CDS in space,
but in terms of cell-averaged data

$$\frac{d \bar{\Phi}}{dt} = \frac{\nu}{\Delta x^2} \mathbf{B}(1, -2, 1) \bar{\Phi} + (\mathbf{bc})$$

Expressing fluxes at the surface based on cell-averaged (nodal) values: Summary of Two Approaches and Boundary Conditions

up of surface/volume integrals: 2 approaches (do things in opposite order)

- (i) Evaluate integrals using classic rules (symbolic evaluation); (ii) Then, to obtain the unknown symbolic values, interpolate based on cell-averaged (nodal) values

$$\left. \begin{array}{l} (i) F_e = \int_{S_e} f_\phi dA \Rightarrow F_e = \mathcal{G}(\phi_e) \\ (ii) \phi_e = \mathcal{H}(\bar{\phi}_P 's) \equiv \mathcal{H}(\phi_P 's) \end{array} \right\} \Rightarrow F_e = \mathcal{F}(\bar{\phi}_P 's)$$

Similar for other integrals:
 $(S_\phi = \int_V s_\phi dV, \bar{\Phi} = \frac{1}{V} \int_V \rho \phi dV, etc)$

- (i) Select shape of solution within CV (piecewise approximation); (ii) impose volume constraints to express coefficients in terms of nodal values; and (iii) then integrate. (this approach was used in the examples).

$$\left. \begin{array}{l} (i) \phi_{a_i}(x) \equiv \mathcal{J}_{a_i}(x) \\ (ii) \int_{V_P} \phi_{a_i}(x) \equiv \bar{\phi}_P \\ (iii) F_e = \int_{S_e} f_{\phi_P} dA \end{array} \right\} \Rightarrow \phi_{a_i}(x) \equiv \phi_{\bar{\phi}_P}(x) \left\{ \Rightarrow F_e = \mathcal{F}(\bar{\phi}_P 's) \right.$$

Similar for higher dimensions:

$$\phi(x, y) \equiv \mathcal{J}_{a_i}(x, y); \quad etc$$

$$\phi_{a_i}(x_P, y_P) \equiv \phi_P; \quad etc$$

• Boundary conditions:

- Directly imposed for convective fluxes
- One-sided differences for diffusive fluxes

Approach 1: Evaluate integrals symbolically, then interpolate based on neighboring cell-averages

- Surface/Volume integrals: Approach 1
 - (i) Evaluate integrals based on classic rules (symbolic evaluation)
 - (ii) Then, to obtain the unknown symbolic values, interpolate based on neighboring cell-averaged (nodal) values
- If we utilize this approach 1

- Symbolic evaluation:

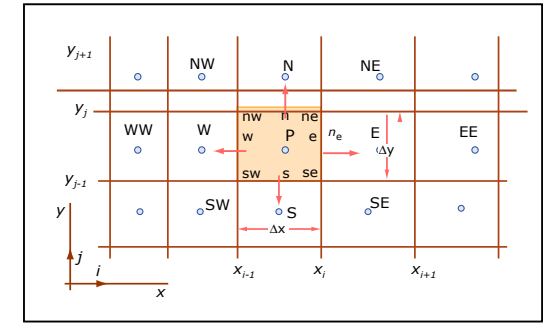
- To evaluate total surface fluxes (convective + diffusive),

$$\int_S \vec{F}_\phi \cdot \vec{n} \, dA = \int_S \rho \phi (\vec{v} \cdot \vec{n}) dA + \int_S \vec{q}_\phi \cdot \vec{n} \, dA$$

values of ϕ and its gradient normal to the cell face at one or more locations on that face are needed. They have to be expressed as a function of nodal values $\bar{\phi}$

- Similar for volume integrals
- Next is interpolation:
 - Express the ϕ 's as a function of nodal values. Numerous possibilities. We already saw some of the most common, provided again next.

Approx. of Surface/Volume Integrals: Classic symbolic formulas



Notation used for a Cartesian 2D and 3D grid.
Image by MIT OpenCourseWare.

• Surface Integrals $F_e = \int_{S_e} f_\phi dA$

– 2D problems (1D surface integrals)

- Midpoint rule (2nd order): $F_e = \int_{S_e} f_\phi dA = \bar{f}_e S_e = f_e S_e + O(\Delta y^2) \approx f_e S_e$
- Trapezoid rule (2nd order): $F_e = \int_{S_e} f_\phi dA \approx S_e \frac{(f_{ne} + f_{se})}{2} + O(\Delta y^2)$
- Simpson's rule (4th order): $F_e = \int_{S_e} f_\phi dA \approx S_e \frac{(f_{ne} + 4f_e + f_{se})}{6} + O(\Delta y^4)$

– 3D problems (2D surface integrals)

- Midpoint rule (2nd order): $F_e = \int_{S_e} f_\phi dA \approx S_e f_e + O(\Delta y^2, \Delta z^2)$
- Higher order more complicated to implement in 3D

• Volume Integrals: $S_\phi = \int_V s_\phi dV$, $\bar{\Phi} = \frac{1}{V} \int_V \rho \phi dV$

– 2D/3D problems, Midpoint rule (2nd order): $S_P = \int_V s_\phi dV = \bar{s}_P V \approx s_P V$

– 2D, bi-quadratic (4th order, Cartesian): $S_P = \frac{\Delta x \Delta y}{36} [16s_P + 4s_s + 4s_n + 4s_w + 4s_e + s_{se} + s_{sw} + s_{ne} + s_{nw}]$

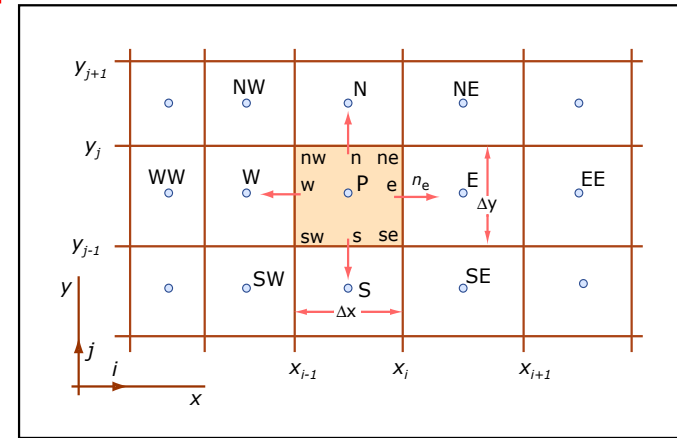
Interpolations and Differentiations

(to obtain fluxes “ F_e ” as a function of cell-average values)

• Upwind Interpolation (UDS) for convective fluxes

- Approximates ϕ_e by its value at the node upstream of “e”. This is equivalent to using backward or forward-difference approx for a first derivative (depends on direction of flow) => Upwind Differencing Scheme, which is also called Donor-cell.

$$\phi_e = \begin{cases} \phi_P & \text{if } (\vec{v} \cdot \vec{n})_e > 0 \\ \phi_E & \text{if } (\vec{v} \cdot \vec{n})_e < 0 \end{cases}$$



Notation used for a Cartesian 2D and 3D grid.
Image by MIT OpenCourseWare.

- This approximation never yields oscillatory solutions (boundedness criterion), but it is numerically diffusive:

- Taylor expansion about x_P : $\phi_e = \phi_P + (x_e - x_P) \frac{\partial \phi}{\partial x} \Big|_P + \frac{(x_e - x_P)^2}{2} \frac{\partial^2 \phi}{\partial x^2} \Big|_P + R_2$
- UDS retains only first term: 1st order scheme in space

$$f_e = \rho \phi_e (\vec{v} \cdot \vec{n})_e \approx \hat{f}_e = \rho \phi_P (\vec{v} \cdot \vec{n})_e \Rightarrow \tau_{\Delta x} = \rho (\vec{v} \cdot \vec{n})_e \Delta x \frac{\partial \phi}{\partial x} \Big|_P + \dots$$

- Leading truncation error is “diffusive”, it has the form of a diffusive flux
- The numerical diffusion is $\rho (\vec{v} \cdot \vec{n})_e \Delta x$ (has 2 components when flow is oblique to the grid)

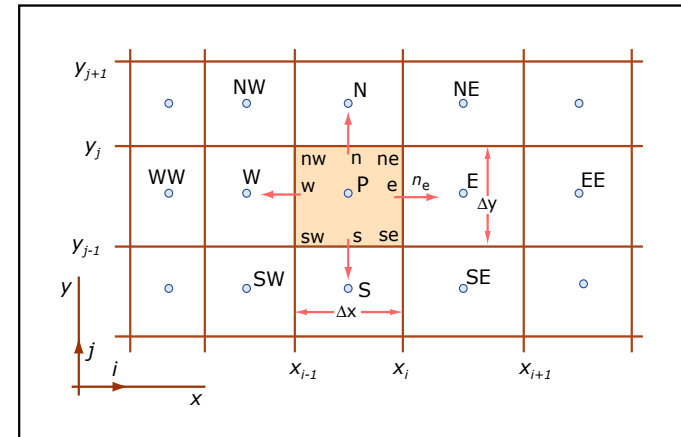
(to obtain fluxes " F_e " as a function of cell-average values)

- Linear Interpolation (CDS) for convective fluxes

- Approximates ϕ_e (value at face center) by its linear interpolation between two nearest nodes:

$$\phi_e = \phi_E \lambda_e + \phi_P (1 - \lambda_e) \quad \text{where } \lambda_e = \frac{x_e - x_P}{x_E - x_P}$$

- λ_e is the interpolation factor



Notation used for a Cartesian 2D and 3D grid.
Image by MIT OpenCourseWare.

- This approx. is 2nd order accurate (for convective fluxes):

- Use Taylor exp. of ϕ_E about x_p to eliminate 1st derivative in Taylor exp. of ϕ_e (previous slide)

$$\begin{aligned} \phi_E &= \phi_P + (x_E - x_P) \frac{\partial \phi}{\partial x} \Big|_P + \frac{(x_E - x_P)^2}{2} \frac{\partial^2 \phi}{\partial x^2} \Big|_P + R_2 \Rightarrow \frac{\partial \phi}{\partial x} \Big|_P = \frac{\phi_E - \phi_P}{x_E - x_P} - \frac{(x_E - x_P)}{2} \frac{\partial^2 \phi}{\partial x^2} \Big|_P - \frac{R_2}{x_E - x_P} \\ \Rightarrow \phi_e &= \phi_P + (x_e - x_P) \frac{\partial \phi}{\partial x} \Big|_P + \frac{(x_e - x_P)^2}{2} \frac{\partial^2 \phi}{\partial x^2} \Big|_P + R_2 = \phi_E \lambda_e + \phi_P (1 - \lambda_e) - \frac{(x_e - x_P)(x_E - x_e)}{2} \frac{\partial^2 \phi}{\partial x^2} \Big|_P + R'_2 \end{aligned}$$

- Truncation error is proportional to square of grid spacing, on uniform/non-uniform grids.
- As all approximations of order higher than one, this scheme can provide oscillatory solutions
- Corresponds to central differences, hence its CDS name (gives avg. if uniform grid spacing)

(to obtain fluxes " F_e " as a function of cell-average values)

- Linear Interpolation (CDS) for diffusive fluxes

- Linear profile between two nearest nodes leads to simplest approx. of gradient (diffusive fluxes)

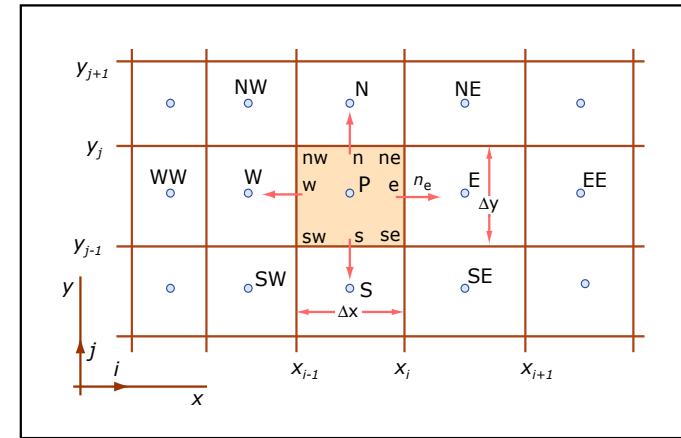
$$\phi = \phi_E \lambda + \phi_P (1 - \lambda) \quad \Rightarrow$$

$$\lambda = \frac{x - x_P}{x_E - x_P}$$

$$\left. \frac{\partial \phi}{\partial x} \right|_e \approx \frac{\phi_E - \phi_P}{x_E - x_P}$$

- Taylor expansions of ϕ 's around x_e , one obtains:

$$\tau_{\Delta x} = \frac{(x_e - x_p)^2 - (x_E - x_e)^2}{2(x_E - x_p)} \frac{\partial^2 \phi}{\partial x^2} \bigg|_e - \frac{(x_e - x_p)^3 + (x_E - x_e)^3}{6(x_E - x_p)} \frac{\partial^3 \phi}{\partial x^3} \bigg|_e + R_3$$



Notation used for a Cartesian 2D and 3D grid.
Image by MIT OpenCourseWare.

- Approximation is 2nd order accurate if e is midway between P and E (e.g. uniform grid)
- When the grid is non-uniform, the formal accuracy is 1st order, but error reduction when grid is refined is asymptotically 2nd order

Interpolations and Differentiations

(to obtain fluxes “ F_e ” as a function of cell-average values)

• Quadratic Upwind Interpolation (QUICK), convective fluxes

- Approx. by quadratic profile between two nearest nodes.
- In accord with convection, third point chosen on upstream side:
 - i.e. chose W if flow is from P to E, or EE if flow from E to P.

This gives:

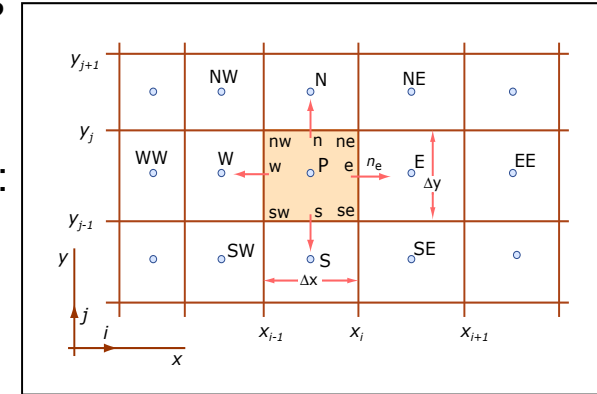
$$\phi_e = \phi_U + g_1 (\phi_D - \phi_U) + g_2 (\phi_U - \phi_{UU})$$

where D, U and UU denote the downstream, first upstream and second upstream, respectively

- Coefficients in terms of nodal coordinates: $g_1 = \frac{(x_e - x_U)(x_e - x_{UU})}{(x_D - x_U)(x_D - x_{UU})}$; $g_2 = \frac{(x_e - x_U)(x_D - x_e)}{(x_U - x_{UU})(x_D - x_{UU})}$
- Uniform grids: coefficients of ϕ 's are 3/8 for node D, 6/8 for node U and -1/8 for node UU
- Somewhat more complex scheme than CDS (larger computational molecules by one node in each direction)
- Approximation is 3rd order accurate on both uniform and non-uniform grids. For uniform grids:

$$\phi_e = \frac{6}{8}\phi_U + \frac{3}{8}\phi_D - \frac{1}{8}\phi_{UU} - \frac{3\Delta x^3}{48} \frac{\partial^3 \phi}{\partial x^3} \bigg|_U + R_3$$

- But, when this interpolation scheme is used with midpoint rule for surface integral, becomes 2nd order



Notation used for a Cartesian 2D and 3D grid. Image by MIT OpenCourseWare.

Interpolations and Differentiations

(to obtain fluxes “ $F_e = f(\phi_e)$ ” as a function of cell-average values)

Higher Order Schemes (for convective/diffusive fluxes)

- Interpolations of order higher than 3 make sense if integrals are also approximated with higher order formulas
- In 1D problems, if Simpson’s rule (4th order error) is used for the integral, a polynomial interpolation of order 3 can be used:

$$\phi(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

(Note: higher-order, approach 1 $\rightarrow$ approach 2 !)

$\Rightarrow$ 4 unknowns, hence 4 nodal values (W, P, E and EE) needed

= Symmetric formula for ϕ_e : no need for “upwind” as with 0th or 2nd order polynomials (donor-cell & QUICK)

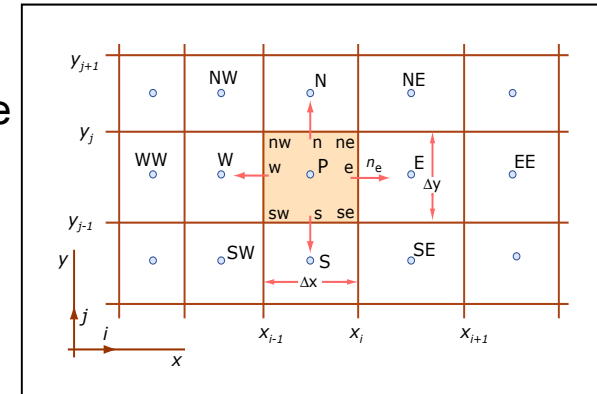
- With $\phi(x)$, one can insert $\phi_e = \phi(x_e)$ in symbolic integral formula. For a uniform Cartesian grid:

- Convective Fluxes:
$$\phi_e = \frac{27\phi_P + 27\phi_E - 3\phi_W - 3\phi_{EE}}{48}$$
 (similar formulas used for ϕ values at corners)

- For Diffusive Fluxes (1st derivative):

$$\left. \frac{\partial \phi}{\partial x} \right|_e = a_1 + 2a_2x + 3a_3x^2 \quad \Rightarrow \quad \text{for a uniform Cartesian grid: } \left. \frac{\partial \phi}{\partial x} \right|_e = \frac{27\phi_E - 27\phi_P + \phi_W - \phi_{EE}}{24 \Delta x}$$

- This FV approximation often called a 4th-order CDS (linear poly. interpol. was 2nd-order CDS)
- Polynomials of higher-degree or of multi-dimensions can be used, as well as cubic splines (to ensure continuity of first two derivatives at the boundaries). This increases the cost.



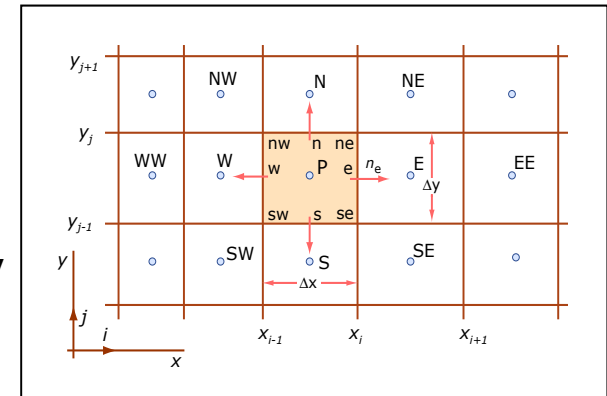
Notation used for a Cartesian 2D and 3D grid.
Image by MIT OpenCourseWare.

Interpolations and Differentiations

(to obtain fluxes “ $F_e = f(\phi_e)$ ” as a function of cell-average values)

Higher Order Schemes

- Polynomial of higher order lead too large computational molecules => use deferred-correction schemes and/or compact (Pade’) schemes
- Ex. 1: obtain the coefficients of $\phi(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ by fitting two values and two 1st derivatives at the two nodes on either side of the cell face. With evaluation at x_e :



Notation used for a Cartesian 2D and 3D grid.
Image by MIT OpenCourseWare.

- 4th order scheme:
$$\phi_e = \frac{\phi_P + \phi_E}{2} + \frac{\Delta x}{8} \left(\frac{\partial \phi}{\partial x} \Big|_P - \frac{\partial \phi}{\partial x} \Big|_E \right) + O(\Delta x^4)$$

- If we use CDS to approximate derivatives, result retains 4th order:

$$\phi_e = \frac{\phi_P + \phi_E}{2} + \frac{\phi_P + \phi_E - \phi_W - \phi_{EE}}{16} + O(\Delta x^4)$$

- Ex. 2: use a parabola, fit the values on either side of the cell face and the derivative on the upstream side (equivalent to the QUICK scheme, 3rd order)

$$\phi_e = \frac{3}{4} \phi_U + \frac{1}{4} \phi_D + \frac{\Delta x}{4} \frac{\partial \phi}{\partial x} \Big|_U$$

- Similar schemes are obtained for derivatives (diffusive fluxes), see Ferziger and Peric (2002)

Other Schemes: more complex and difficult to program

- Large number of approximations used for “convective” fluxes: Linear Upwind Scheme, Skewed Upwind schemes, Hybrid. Blending schemes to eliminate oscillations at higher order.

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2.29 Numerical Fluid Mechanics

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